



Week 1: Recursion



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Outline



Recursion

- Components of Recursion
- Summation
- Exponentiation
- Factorial



What is Recursion?



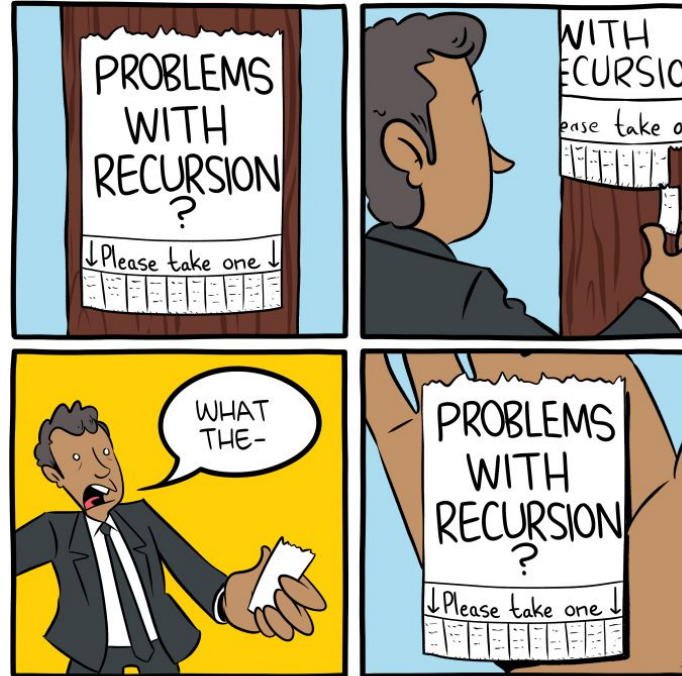
- It breaks down a large problem into smaller subproblems that are simpler to solve^[1].
- A function that is calling itself one or more times^[2].

[1] Bradley N. Miller, David L. Ranum, Problem Solving with Algorithms and Data Structures using Python, 2011

[2] Michael T. Goodrich et al., Data Structures and Algorithms in Python, 2013



What is Recursion?



smbc-comics.com

Recursion



Real-life example:

- Fractal patterns in art and nature
- **Matryoshka** doll
 - Also known as Russian doll or nested doll
 - Set of wooden dolls where smaller dolls are placed inside another.
 - A symbol of motherhood and fertility.





When to use recursion?



- For problems that contain smaller instances of the same problem.
- To repeat a computer program, **while-loop** and **for-loop** can be used.
- Alternatively, **recursion** repeats by calling a function itself one or more times.



Components of Recursion



- Base Case: The smallest subproblem that can be solved easily.
- Recursive Case: A subproblem that reduces the size of the input and move toward the base case



Summation



- Summing list elements recursively.



Summation



- Summing list elements recursively.

Time Complexity: $O(?)$

```
def getSum(data):  
    """ Return the sum of the first  
    n numbers of sequence data."""  
    totalSum = 0  
    for i in data:  
        totalSum = totalSum + i  
    return totalSum
```

```
def getSum(data):  
    """ Return the sum of the first n  
    numbers of sequence data."""  
    if len(data)==0:  
        return 0  
    else:  
        return data[0]+  
        getSum(data[1:])
```



Power



Power function takes 2 numbers as input and return the value of *base* to the power of *exp*.

$$\text{power}(\text{base}, \text{exp}) = \begin{cases} 1 & \text{if exp} = 0 \\ \text{base} * \text{power}(\text{base}, \text{exp}-1) & \text{otherwise} \end{cases}$$

```
def getPower(base,exp):  
    """ Return the multiplciation of base with exp times. """  
    if exp==0:  
        return 1  
    else:  
        return base*getPower(base, exp-1)
```



Factorial



Factorial

- Used to calculate the number of **permutations** of n items.
- That is, given n items, how many ways to arrange them into a sequence.
- For instance, 3 characters: 1, 2, 3 can be arranged in $3! = 3*2*1 = 6$ ways.
 - 123, 132, 213, 231, 312, 321



Factorial



Factorial function takes a number as an input and returns the factorial of that number

- $n!$ = Product of positive integers from 1 to n
- If $n = 0$, then $n! = 1$ by convention

factorial(1)

factorial(2)

factorial(3)

factorial(4)

factorial(5)



Factorial



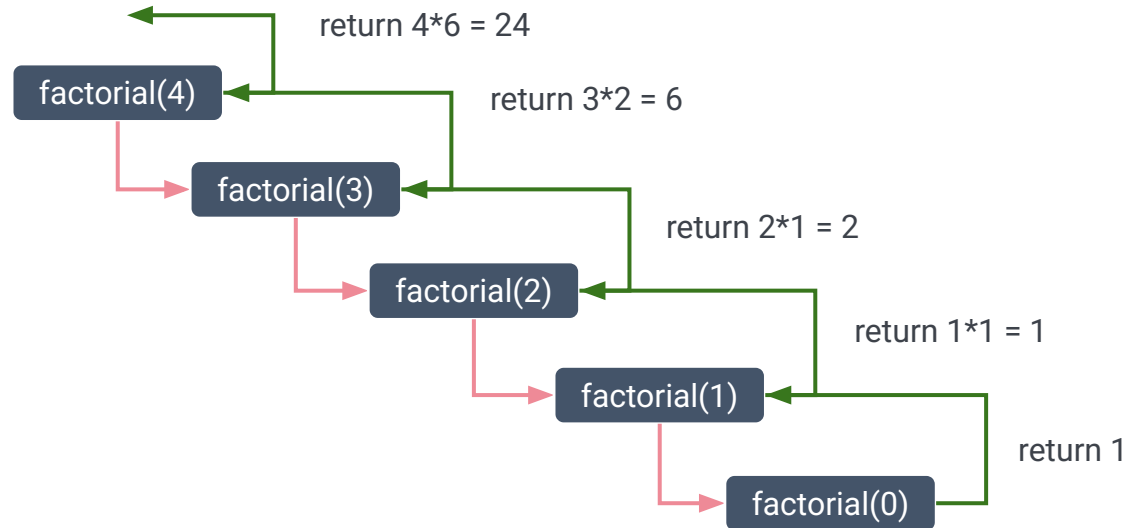
Factorial function takes a number as an input and returns the factorial of that number

- $n!$ = Product of positive integers from 1 to n
- If $n = 0$, then $n! = 1$ by convention

Recursion



- For recursive, repetition is conducted by repeatedly invoke the function call.





Factorial



Factorial function takes a number as an input and returns the factorial of that number

- $n!$ = Product of positive integers from 1 to n
- If $n = 0$, then $n! = 1$ by convention
- Formally, for any integers $n \geq 0$,

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ n * (n - 1) * (n - 2) * \dots * 3 * 2 * 1 & \text{if } n \geq 1 \end{cases}$$



Recursion



- Recursive definition of factorial function

$$n! = \begin{cases} 1 & \text{if } n = 0 \quad (\text{base case}) \\ n * (n - 1)! & \text{if } n \geq 1. \quad (\text{recursive case}) \end{cases}$$

- For example, $4! = 4 * 3!$



Recursion



Loop

```
# Factorial loop version
def factorial1(n):
    if n == 0:
        return 1
    else:
        total = 1
        for num in range(2, n+1):
            total *= num
        return total
```

Time Complexity: $O(?)$

Recursive

```
# Factorial recursive
version
def factorial2(n):
    if n == 0:
        return 1
    else:
        return
n*factorial2(n-1)
```

Time Complexity: $O(?)$



Recap



- Components:
 - Base case and Recursive case
- Examples
 - Summation
 - Power
 - Factorial
 - Binary search
- One rule for recursion: there must be a stop condition.